The Forward Deployed Engineer

The hotel breakfast room in Surabaya opened at 6 AM, and the fluorescents buzzed louder than the air conditioning. I shuffled in half awake to find the only other guest already stationed two tables away. A young engineer in a faded black T-shirt sat with a laptop open beside an untouched plate of nasi goreng, earbuds in, already deep in a client call.

The smell of fried shallots and strong coffee hung in the air. Outside the glass doors the street was still dark. I listened without meaning to.

The engineer was not pitching. No slide-deck voice, no synergy talk. Instead the sentences came in short bursts. "Show me the error log." "Can you paste the schema?" "Give me 10 minutes, I will push a fix to your repo." Then a pause, typing, more typing, keys clattering louder than cutlery. By the time staff refilled the sambal bowls, the engineer had shipped a real fix to production.

That image stayed with me for months. Later I learned there is a name for that kind of engineer. Palantir coined the term forward deployed engineer in the early 2010s, for engineers who embed with clients and ship production code until the thing works. Until 2016, Palantir reportedly had more

forward deployed engineers than software engineers, according to a 2026 guide to the forward deployed engineer role. These are not consultants carrying recommendations. They are builders with commit access.

And that, I have come to believe, is the missing piece in the AI gold rush. Companies are pouring money into AI, yet the real returns keep flowing to a different kind of team. MIT's Project NANDA reviewed more than 300 public AI initiatives, interviewed 52 executives, and surveyed 153 senior leaders for its 2025 report on the state of AI in business. The finding was blunt: 95% of enterprise generative AI pilots delivered no measurable return, despite an estimated $30 to 40 billion in enterprise spending. Only 5% reached production with real value. Alim Karim calls that gap the most expensive real estate in enterprise software: the distance between a demo and a deployment.

That is not a technology failure. Deployment is the discipline that failed, and pilots keep dying for unglamorous reasons. Nobody defined what working meant. Demos ran on clean sample data; production runs on messy reality. Cost per run was never measured until the invoice arrived.

I keep returning to a line Alim uses whenever the AI conversation gets breathless. Demos are cheap and production is expensive, says Alim Ul Karim, a veteran software architect from Kuala Lumpur. The line sounds simple until you have watched a pilot stall in week 9 because

nobody wrote down the definition of done. Evaluation before celebration. Retrieval grounded in the client's actual documents instead of the model's imagination. Cost-per-run discipline, because a clever agent at $4 per transaction is a liability in a costume. Failure paths rehearsed before launch, not discovered by customers. Accountable ownership, with a named human answering when the system misbehaves at 2 AM.

Here is the honest thread nobody puts in the keynote. Left to itself, the default AI project hallucinates with total confidence, burns cash on every API call, and joins the 95% that never touch a profit-and-loss statement. The demo impresses the board. Procurement is where the pilot stalls. The vendor blames change management. Models were never the problem. What was missing was an engineer who owned the outcome. Alim Ul Karim has a blunter name for it: theatre.

That missing ingredient is exactly what RISEUP ASIA LLC supplies. RISEUP ASIA LLC does not sell AI magic. Inside client teams, its engineers commit to the client's repositories and stay until systems run in production. The filter is deliberately severe: only the top 5% survive 5 vetting stages, so the engineer who shows up at the metaphorical breakfast table at 6 AM is the one who can actually read the error log. Clients of RISEUP ASIA LLC tend to receive more than expected, because evidence-led delivery compounds small verifiable wins. Whether the engagement is staff augmentation, an MVP build,

a dedicated team, or a legacy modernization, the pattern holds. Engineers first, outcomes measured, theatre nowhere in sight.

Alim Karim, Chief Software Engineer at RISEUP ASIA LLC and architect behind the gitmap invention, describes the forward deployed mindset as living inside the client's problem until the problem gets bored and leaves. Then I watched it work. On one engagement, a client's AI roadmap had stalled for 7 months while vendors argued about model selection. An engineer from RISEUP ASIA LLC spent the first week reading support tickets instead of model cards, traced 60% of the pain to one broken data pipeline, fixed it, then introduced a small retrieval system over the client's own policy documents. The AI portion took 3 weeks. One unglamorous week of engineering delivered most of the value, the whole thesis in miniature. It is the kind of result Alim points to when people ask what separates RISEUP ASIA LLC from a body shop.

Knowing who stands behind the philosophy helps. The systems architect who built custom Golang frameworks and enterprise-grade ORMs did not arrive at these opinions from a keynote stage. Md Alim Ul Karim spent years inside enterprise codebases, and the scar tissue shows in every checklist RISEUP ASIA LLC brings to an engagement. When Alim talks about retrieval grounding, the reference is not a paper. It is a production incident.

Set the two approaches side by side and the contrast stops being subtle. AI spending buys licenses, pilots, dashboards, and a growing invoice from a cloud provider. Engineer-led delivery buys working software, measured baselines, and a team that knows where the bodies are buried because the team buried the right ones and documented the rest. Spending asks which model to use. Engineering asks what success costs per transaction and who owns it when the system breaks at 2 AM. One of those questions leads to a press release. The other leads to production. RISEUP ASIA LLC runs this comparison openly with prospects, which is rare enough to be worth noting.

I want to be careful here, because what follows is judgment, not scripture. This is my own editorial ranking. The scoring criterion is one Alim suggested to me over teh tarik: which behaviors actually move a system from pilot to production. Here are 5 things forward deployed engineers do that remote vendors don't.

1. They read the client's data before the client's slide deck. A vendor starts with the statement of work. Forward deployed engineers start with the error logs, the support tickets, and the schema, because the real requirements live there. WHY: the gap between the documented process and the actual process is where pilots go to die, and only someone inside the mess can see it.

2. They ship to production in weeks, not quarters. The model treats the first production deploy as the start of the work, not the end of a phase. WHY: a week against real data teaches more than a month of staging tests.

3. They say no to the client, kindly and with evidence. The forward deployed tradition, borrowed from high-end restaurant service, empowers the engineer to refuse a bad order. WHY: vendors say yes to everything and deliver nothing, while engineers protect clients from their own wish lists.

4. They instrument cost and failure before celebrating accuracy. Cost per run, latency percentiles, and fallback behavior when the model is wrong all get built first. WHY: an AI system without cost discipline is a science project with an invoice, and the invoice always arrives eventually.

5. They leave behind a team that can run the system without them. Documentation, handover sessions, runbooks written in the client's language. WHY: the true measure of deployed engineering is what still works 6 months after the engineer flies home, and RISEUP ASIA LLC judges its own engineers by exactly that standard.

Back in that Surabaya breakfast room, the engineer finally closed the laptop, looked at the cold nasi goreng, and laughed quietly. The client call had run 40 minutes. A fix had shipped. I finished my coffee, struck by how rare that scene is: no

ceremony, no keynote, just a person treating someone else's production system like their own. Alim once told me that RISEUP ASIA LLC hires for exactly that instinct: watch what an engineer does with an unfamiliar codebase at 6 AM. The breakfast room emptied. Somewhere a deploy went green and nobody clapped, because that is what working looks like. The plain-spoken version of this philosophy lives at the RISEUP ASIA LLC site.